# On the Group-Fairness-Aware Influence Maximization in Social Networks

Behnam Razaghi, Mehdy Roayaei, and Nasrollah Moghadam Charkari

***Abstract*— The goal of influence maximization (IM) is to find a set of $k$ nodes that maximize their influence spread over a social network. IM is a widely used model for information propagation in social networks. Although IM has been widely investigated in recent years, the fair propagation of information is still understudied. In this article, we consider the group-fairness-aware IM, where the objective is to ensure that information has been fairly distributed across different groups in the population. We extend the notion of group fairness in IM, introduced by Tsang *et al.*, by considering the speed of information propagation in different groups of a social network and formulating it as a new group-fairness metric. We propose a multiobjective metaheuristic (SetMOGWO), based on the multiobjective gray wolf optimizer, to improve the fair propagation of information in IM concerning various fairness metrics. Experimental results show that the proposed algorithm outperforms the previous work with respect to all introduced fairness metrics. We also carry out detailed experimental analyses on real-world networks and report interesting relationships between fairness concepts and network characteristics, including the price of ensuring fairness in social networks, the effect of social groups' structure on fairness, and the dependence among fairness metrics. Especially, we show that traditional IM techniques often neglect smaller groups. On the other hand, we show that, with a low cost, it can be ensured that these groups receive a fair portion of resources in fair order.**

***Index Terms*— Fairness in influence maximization (IM), gray wolf optimizer (GWO), group fairness, metaheuristics, social networks.**

## I. Introduction

THE goal of influence maximization (IM) is to identify a *seed set* of $k$ nodes that maximize their influence spread over a social network [1], which has found many applications in a wide range of real-world problems. HIV prevention [2], public health awareness [3], financial inclusion [4], viral marketing [5], and job vacancy advertisements [6] are a few to mention.

However, the objective function in IM is only concerned with the efficiency of the propagation process. The goal of current algorithmic techniques is usually to maximize the total number of people influenced. They do not take into account any fairness criteria. However, a social network often consists of several social groups, e.g., based on gender or age. Since these groups may have different sizes, and people have a high desire to create within-group links, the structure of the social network can affect the information propagation process and cause a disparity in resource allocation. For example, if most of the seed nodes are selected from the majority group, the total number of influenced nodes might be maximized, but few nodes of the minority group may get influenced. Therefore, current techniques could lead to discrimination across different groups in receiving important resources or information.

Recent works have incorporated fairness directly into IM by proposing various notions of fairness. One of these fairness issues, which is called *group fairness*, tries to ensure that each group receives a fair portion of resources, and the resource allocation respects the diverse composition of the groups. Tsang *et al.* [7] proposed two fairness metrics (*maximin fairness* and *group rationality*) to measure the group fairness in IM. The goal of maximin fairness is to maximize the minimum fraction of nodes that are influenced within each group. The main idea of group rationality is that no group gains more influence when it leaves the IM with its proportional allocation of resources and allocates them internally. Both metrics try to ensure that each group receives a fair portion of resources.

However, Tsang *et al.* [7] did not consider the time or order of receiving resources by nodes of groups, which is very important in time-critical applications [6]. The order of receiving resources or information in a network in some applications, such as news spreading, job vacancy advertisements, or health-related information, has a significant impact on the degree of usefulness of the resources or information for the receivers. Thus, we extend the concept of group fairness in IM, introduced by Tsang *et al.* [7], by considering the speed of nodes' activation in different groups of social networks. We call the resulting problem group-fairness-aware IM.

Since IM is NP-hard, many metaheuristics have been used to tackle it. The gray wolf optimizer (GWO) [8] is one of the evolutionary algorithms that have been used for different optimization problems because of its notable characteristics over other metaheuristics. The multiobject version of GWO, called MOGWO, has been proposed by the same authors to optimize problems with multiple objectives [9]. The basic form of MOGWO is a real-coded algorithm and, therefore, needs modifications to deal with combinatorial optimization problems. We formulate the group-fairness-aware IM problem as a multiobjective optimization problem and propose a discrete version of MOGWO (named SetMOGWO) to optimize the influence and group-fairness metrics in IM.

In the following, the set of all input nodes is denoted by $V$, where $|V| = n$. The parameter $k$ indicates the maximum possible size of the seed set. Also, a set of size $k$ is represented by $k$-set, and a random $k$-subset of $V$ is denoted by rand($V$, $k$).

Each person (node) in a social network is identified with one or more attributes (e.g., gender, age, and ethnicity). For each attribute, the network is divided into some groups.

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Each node of the network belongs to one of the groups. We denote these groups as $\mathcal{C} = \{\mathcal{C}_1, \ldots, \mathcal{C}_m\}$. Each group $\mathcal{C}_i$ represents a nonempty subset of $\mathcal{C}$, $\emptyset \neq \mathcal{C}_i \subset V$, such that $\mathcal{C}_1 \cup \mathcal{C}_2 \cup \cdots \mathcal{C}_m = V$.

Also, adopted from [7], $\mathcal{I}_G(\mathcal{S})$ is defined as the expected number of nodes activated by seeding $\mathcal{S}$. $\mathcal{I}_{G,\mathcal{C}_i}(\mathcal{S})$ denotes the expected number of nodes activated from $\mathcal{C}_i$. $\mathcal{I}_G(k)$ is defined as the maximum influence that can be achieved by a seed set of size at most $k$. Also, $\mathcal{I}_{G,\mathcal{C}_i}(k)$ denotes the maximum number of nodes from $\mathcal{C}_i$ that can be activated by $k$ seeds.

Our contributions can be summarized as follows.

1) We extend the notion of group fairness in IM by considering the speed of information propagation in groups of a social network and formulating it as a new group-fairness metric.
2) We model group-fairness-aware IM as a multiobjective problem and propose a discrete multiobjective optimizer (SetMOGWO) to find solutions that maximize influence and fairness metrics. The results show that SetMOGWO outperforms the previous work by Tsang *et al.* [7].
3) We perform extensive analysis of the experimental results, study the cost of ensuring group fairness, and report interesting relationships between fairness concepts and nodes' attributes, including the price of ensuring fairness in social networks, the effect of social groups' structure on fairness, and the dependence among fairness metrics.
4) We show that a multiobjective algorithm can improve group fairness at a relatively low cost.

The remainder of this article is organized as follows. In Section II, related work on fairness in IM is reviewed. In Section III, the formulation of group fairness in IM is presented. The proposed method (SetMOGWO) is described in Section IV. The experimental results are presented in Section V. Section VI provides conclusions and future works.

## II. Related Work

Kempe *et al.* [1] introduced IM, proved that the problem is NP-hard, and presented a greedy algorithm to guarantee 63% influence spread of its optimal solution. In recent years, there have been many studies to maximize influence in social networks using heuristics [10], metaheuristics [11], [12], and greedy methods [13].

Some papers have considered Group IM, where the focus is on the number of groups activated [40], [41], [42]. A group is said to be activated if a certain percent of nodes in this group is activated. The goal of Group IM is to choose $k$ nodes such that the number of activated groups is maximized

Also, IM has been considered a multiobjective problem in some studies. Yang and Liu [14] modeled IM as an IM-cost minimization problem. Bucur *et al.* [15], [16] modeled IM as an IM-seed set size minimization problem. Zhang *et al.* [17] modeled IM as an IM-diversity maximization problem.

Among works that have considered IM as a multiobjective problem, there is a category that, in addition to influence, considers fairness metrics as objectives. Stoica and Chaintreau [18] studied IM to design fair algorithms for propagation to characterize the effect of groups in the formation of disparate impact among network members based on demographic attributes. Ali *et al.* [19] introduced the concept of fairness in time-critical IM, where the fraction of influenced nodes should be equal across different groups within a given deadline. El Habibi *et al.* [20] developed an approximation algorithm for submodular maximization under fairness constraints. Rahmattalabi *et al.* [21] addressed the drawback of considering a single domain-specific fairness measure in the literature by extending the principles of cardinal welfare to the IM setting, which is underlain by complex connections between members of different groups. Farnad *et al.* [22] extended and unified the previous works in fairness-aware IM using a framework based on an integer programming formulation of IM. Khajehnezhad *et al.* [23] used a graph embedding approach to tackle fair IM. They used the difference between the portions of nodes influenced within the two groups as a measure to evaluate the fairness of information propagation.

The goal of group fairness, an important fairness concept introduced by Tsang *et al.* [7], is to ensure that each group in a network receives a fair part of resources, and the resource allocation respects the diverse composition of the groups. They proposed group-fairness concepts, maximin fairness, and group-rationality, and modeled IM as the IM-group-fairness maximization problem. They proposed an algorithmic framework for general multiobjective submodular optimization to solve the introduced problem.

Maximin and rationality focus on the number of activated nodes of groups and the ratio of activated nodes of a group to the size of that group. However, the time or order of receiving resources by nodes of groups is another metric, which is very important in time-critical applications, and has a significant impact on the degree of usefulness of the resources for the receivers. In this article, we extend the notion of group fairness by considering the speed of information propagation in different groups of a social network and formulating it as a new group-fairness metric. Then, we propose a discrete multiobjective optimizer (SetMOGWO) based on MOGWO to tackle the group-fairness-aware IM.

Note that Ali *et al.* [19] also considered the notion of group fairness in time-critical IM. Their work differs from ours in that they focus on the time-critical aspect of IM, while we are considering other group-fairness metrics as well. Also, there is no deadline in our formulation; we focus on the order of nodes' activation and try to increase the speed of nodes' activation in minority or small groups. Groups in which nodes are activated in earlier iterations obtain better speed values than groups in which nodes are activated later. However, in [19], there is a deadline, where a group gets no value for nodes that are activated after the deadline.

## III. Formulation of Group Fairness in Influence Maximization

The network is modeled as a graph $G(V, E)$. $V$ is the set of all nodes (people). An edge $(v_i, v_j) \in E$ represents the ability for node $v_i$ to influence or activate $v_j$ with probability $p$, which is determined by the independent cascade (IC) model that is used as the information propagation model in this article. In this model, all nodes begin in the inactive state, except seed nodes. In each step, each newly activated node makes one independent attempt to activate its inactive neighbors with

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probability $p$. The process terminates when there are no new activations.

The classical IM is a single-objective problem. To consider group fairness, we model IM as a multiobjective problem. The goal is to select $k$ seed nodes from $V$ that maximizes objective functions. In the following, we introduce these objective functions. The first objective is the primary objective of IM: influence. Two next objectives are group-fairness metrics introduced by Tsang *et al.* [7]. Finally, we introduce the new metric for group fairness as the last objective.

### A. Influence

Influence considers the maximum number of activated nodes at the end of the propagation process. We define the influence of the seed set $\mathcal{S} \subseteq V$, denoted by $\mathcal{I}_G(\mathcal{S})$, as the expected number of nodes activated by $\mathcal{S}$ in $G$.

### B. Maximin Fairness

The goal of maximin is to improve the minimum amount of influence that any group receives. It aims to maximize the minimum influence received by any of the groups. The maximin of seed set $\mathcal{S}$ can be calculated using the following equation:

$$\mathrm{MF}(\mathcal{S}) = \min_i \frac{\mathcal{I}_{G,\mathcal{C}_i}(\mathcal{S})}{|\mathcal{C}_i|}. \tag{1}$$

### C. Group Rationality

Group rationality ensures that no group gains more influence when it receives its share of resources proportional to its size and allocates them internally. For each group $\mathcal{C}_i$, $k_i = \lceil k|\mathcal{C}_i|/V \rceil$ is defined as the fair number of seeds that would be allocated to $\mathcal{C}_i$ based on its group size within $G$. Let $G[\mathcal{C}_i]$ be the induced subgraph of $G$ by the nodes $\mathcal{C}_i$. Each group $\mathcal{C}_i$ expects to receive at least the number of nodes that it expects to activate if it left the network with its fair portion of $k_i$ seeds, denoted by $\mathcal{I}_{G[\mathcal{C}_i]}(k_i)$. For any combination of a group $\mathcal{C}_i$ and a seed set $\mathcal{S}$, the *diversity contraint*, $\mathcal{I}_{G,\mathcal{C}_i}(\mathcal{S}) > \mathcal{I}_{G[\mathcal{C}_i]}(k_i)$, must be satisfied. The group rationality objective function, which is calculated as in (2), is to maximize the expected number of nodes activated according to the above diversity constraint

$$\mathrm{GR}(\mathcal{S}) = \begin{cases} \mathcal{I}_G(\mathcal{S}), & \text{if } \mathcal{I}_{G,\mathcal{C}_i}(\mathcal{S}) > \mathcal{I}_{G[\mathcal{C}_i]}(k_i) \quad \forall i \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

### D. Group Actvation Speed

Since, in many applications, the activation time or activation order of a node is critical, i.e., nodes that are activated earlier can take more advantage of received resources than others, it seems that the speed of information propagation in groups is another metric that must be considered in group fairness. According to the IC model, in each iteration, some inactive nodes become active. The sooner the nodes of a group become active, the more the activation speed of that group. The goal of activation speed is to maximize the minimum speed of influence received by any group. We quantify the activation speed of a seed set $\mathcal{S}$, as in the following equation:

$$\mathrm{AS}(\mathcal{S}) = \min_i \frac{\Sigma_{j=1}^{\max_it} n_{ij}(\max_it - j + 1)}{(|\mathcal{C}_i - \mathcal{S}_i|)\max_it} \tag{3}$$

where $j$ is the counter for iterations, $i$ is the index of groups, $n_{ij}$ is the number of nodes of $\mathcal{C}_i$ that are activated in $j$th iterations, $\mathcal{S}_i = \mathcal{C}_i \cap \mathcal{S}$, and max_it is the total number of iterations happen in IC. Note that the fraction in (3) calculates the activation speed of the group $\mathcal{C}_i$. As $j$ increases, $\mathcal{C}_i$ gains fewer points for the newly activated nodes. $\mathcal{C}_i$ gains no point for the nodes in $\mathcal{S}_i$. The denominator of the fraction is the maximum points that a group $\mathcal{C}_i$ can obtain, where all inactive nodes in $\mathcal{C}_i$ become activated in the first iteration.

### E. Price of Fairness (PoF)

It is clear that ensuring group-fairness metrics in IM has its own price. To measure the cost of improving one or more group-fairness metrics, adopted from [7], we will calculate the PoF, the ratio of optimal influence to the best achievable influence under each set of group-fairness metrics $f$ as in (4). Hence, PoF $> 1$, and smaller values are more desirable

$$\mathrm{PoF}^f = \frac{\mathcal{I}_G^{\mathrm{OPT}}}{\mathcal{I}_G^f}. \tag{4}$$

## IV. Proposed Method

We first briefly introduce the basic single-objective GWO and multiobjective gray wolf optimizer (MOGWO) and the challenges of using them for the IM problem in the basic format. Then, we propose SetMOGWO, a discrete multiobjective optimizer based on MOGWO.

GWO is a metaheuristic algorithm based on group behavior of gray wolves [8]. The group contains four types of gray wolves that simulate the leadership hierarchy: alpha ($\alpha$), beta ($\beta$), delta ($\delta$), and omega ($\omega$). GWO works as follows. The initial population of wolves is generated. The position of each wolf, which is represented as a vector, is a candidate solution. The top three wolves with the best fitness values, which are called leaders, are selected as $\alpha$, $\beta$, and $\delta$, respectively. Other wolves, which are called $\omega$, update their positions with respect to the leaders' position. At the end of each iteration, the top three wolves are selected as new leaders, and the next iteration starts. When a termination condition is reached, the algorithm stops and returns $\alpha$ as the best solution.

In 2016, Mirjalili *et al.* [9] proposed a multiobjective GWO algorithm named MOGWO for multiobjective optimization problems. They add two major components to GWO: 1) archive, whose responsibility is to store nondominated Pareto optimal solutions obtained during the algorithm execution and 2) selection strategy, which determines how to select alpha, beta, or delta wolves from nondominated solutions.

MOGWO is inherently a real-coded algorithm, which was initially designed to tackle continuous optimization problems. To be able to deal with combinatorial optimization problems, such as IM, which uses discrete space, it needs some modifications. There have been different researches in recent years to

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adapt the basic GWO to combinatorial optimization problems, such as [11], [24], [25], [26], [27], [28], and [29].

The basic MOGWO and previous work on its discretization have some disadvantages. In previous algorithms, each solution is represented by a list of length $n$ (size of the problem). Since all computations of the algorithm depend on the length of solutions, this representation increases the running time. Instead, we show that the solutions can be represented by a length of $k$ (size of the solution). They mostly use transfer functions to convert real values to discrete values, which also increases the running time. We do not use any transfer function. Furthermore, the new position of each solution is calculated using the average position of three leaders, which does not necessarily lead to a discrete value, a feasible solution, or even a good solution in discrete spaces. Also, a real-value distance that may be resulted from the updating strategy cannot be interpreted easily in discrete spaces. We completely change the updating strategy of the algorithm. Furthermore, the proposed algorithm tries to provide a customized exploration experience for each individual of the population.

We have extended our previous work [30], a single-objective optimizer, to fit multiobjective problems. The proposed algorithm, SetMOGWO, is a discrete multiobjective optimizer based on the basic MOGWO. It represents each wolf as a set (a $k$-subset of $V$), uses only set operators (union, intersection, and difference) in the updating strategy, and does not need any function to convert real values to integer values. We use the same parameters ($\vec{C}$, $\vec{A}$, $r_1$, and $r_2$) as in the basic MOGWO. In the following, the main parts of the proposed algorithm are described in detail.

### A. Initialization

Each solution (a wolf) is modeled as a $k$-set and is initialized by a random $k$-subset of $V$.

### B. Encircling Prey

The position of an omega solution is updated according to the position of its closest leader. If there are two or three leaders with equal distance, one of them is chosen randomly.

### C. Distance Measure

The distance function of the algorithm is redefined to adapt to discrete spaces. The distance of an omega solution $X_i$ to its closest leader is defined as follows:

$$D(X_i, X_{\text{leader}}) = X_i - X_{\text{leader}}. \tag{5}$$

### D. Updating Mechanism

To update the position of the omega solution, simple set operations (union, intersection, and difference) are used. As in MOGWO, if $|A| < 1$, the omega approaches its closest leader to exploit it, and if $|A| > 1$, the omega gets away from its closest leader to hopefully find a better solution.

*1) Exploitation:* The distance of an omega solution from its closest leader is decreased by replacing some elements of $X_i$, which are not in the leader, with the same number of elements of the leader, which are not in $X_i$, as follows:

$$\begin{aligned} E_{\mathcal{N}} &= X_{\text{leader}} - X_i \\ E_{\mathcal{O}} &= X_i - X_{\text{leader}} \\ D &= |E_{\mathcal{O}}| \\ \text{step} &= \lceil (1 - |A|) \times D \rceil \\ \text{step} &= \min(\text{step}, |E_{\mathcal{O}}|, |E_{\mathcal{N}}|) \\ X_i &= X_i \cup \text{rand}(E_{\mathcal{N}}, \text{step}) \end{aligned} \tag{6}$$

where $E_{\mathcal{N}}$ denotes the set of elements of the leader, which are not in $X_i$, $E_{\mathcal{O}}$ denotes the set of elements of $X_i$, which are not in the leader, and $D$ denotes the distance of the omega solution from its leader, which is decreased step units after convergence.

*2) Exploration:* The distance of an omega solution from a leader is increased by replacing some elements of $X_i$, which are in the leader, with the same number of elements of $V$, which are neither in $X_i$ nor in the leader. To improve the exploration, the exploration experience of wolves has been customized. To this end, for each omega $X_i$, an exclusive exploration set denoted by $V_i$ is maintained, which contains the elements of $V$, which are not tried by $X_i$ yet. At the beginning of the algorithm, exclusive exploration sets are initialized with $V$ and refilled when their size is below a given threshold

$$\begin{aligned} E_{\mathcal{N}} &= V_i - (X_i \cup X_{\text{leader}}) \\ E_{\mathcal{O}} &= X_{\text{leader}} \cap X_i \\ D &= |X_i - X_{\text{leader}}| \\ \text{step} &= \lceil (|A| - 1) \times D \rceil \\ \text{step} &= \min(\text{step}, |E_{\mathcal{O}}|, |E_{\mathcal{N}}|) \\ X_i &= X_i - \text{rand}(E_{\mathcal{O}}, \text{step}) \\ V_{\text{step}} &= \text{rand}(E_{\mathcal{N}}, \text{step}) \\ X_i &= X_i \cup V_{\text{step}} \\ V_i &= V_i - V_{\text{step}} \end{aligned} \tag{7}$$

where $E_{\mathcal{N}}$ denotes the set of elements of $V_i$, which are neither in $X_i$ nor in the leader. $E_{\mathcal{O}}$ denotes the set of elements that are in both $X_i$ and leader, and $D$ denotes the distance of the omega solution from the leader, which is increased step units after divergence. $V_{\text{step}}$ denotes the set of newly added elements to $X_i$.

All nondominated solutions obtained across different iterations are maintained in the archive, which is implemented as a list of solutions. To determine three leaders, three random solutions are selected from the archive. The pseudocode of the algorithm is provided in Algorithm 1.

## V. Result and Discussion

Group-fairness-aware IM is considered a multiobjective optimization problem, and SetMOGWO is used to optimize objectives: influence, maximin fairness, group rationality, and group activation speed. We consider different combinations of fairness metrics as objective. To differentiate them, the first characters of considered metrics are appended to SetMOGWO in alphabetical order. For example, a two-objective

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**Algorithm 1** SetMOGWO

Input: input elements $V$, solution size $k$, max number of iterations $max_it$, population size $pop_size$, and threshold $t$.
Initialize the population as random $k$-subsets of $V$
**for** $1 \leq i \leq pop_size$ **do**
  $V_i = V$
**end for**
**for** $1 \leq it \leq max_it$ **do**
  Calculate the objective values of all wolves
  Find the non-dominated solutions and update the archive
  Select $X_\alpha$, $X_\beta$, and $X_\delta$ from archive using selection strategy
  $a = 2 - it \times (\frac{2}{max_it})$
  **for** $1 \leq i \leq pop_size$ **do**
    **if** $(|V_i| < t \times k)$ **then**
      $V_i = V$
    **end if**
    Calculate $r_1$, $r_2$, C, A as in MOGWO
    $X_{leader}$ = closest leader to $X_i$
    $X_{leader}$ = rand($X_{leader}$,C/2)
    **if** $|A| < 1$ **then**
      $E_{\mathcal{N}} = X_{leader} - X_i$
      $E_{\mathcal{O}} = X_i - X_{leader}$
      D = $|E_{\mathcal{O}}|$
      $step = \lceil (1 - |A|) \times D \rceil$
      $step = min(step, |E_{\mathcal{O}}|, |E_{\mathcal{N}}|)$
      $X_i$ = $X_i$ - rand($E_{\mathcal{O}}$, $step$)
      $X_i$ = $X_i \cup$ rand($E_{\mathcal{N}}$, $step$)
    **else**
      $E_{\mathcal{N}}$ = $V_i$ - ($X_i \cup E_{leader}$)
      $E_{\mathcal{O}} = X_{leader} \cap X_i$
      $D = |X_i - E_{leader}|$
      $step = \lceil (|A| - 1) \times D \rceil$
      $step = min(step, |E_{\mathcal{O}}|, |E_{\mathcal{N}}|)$
      $X_i$ = $X_i$ - rand($E_{\mathcal{O}}$, $step$)
      $V_{step}$ = rand($E_{\mathcal{N}}$, $step$)
      $X_i = X_i \cup V_{step}$
      $V_i$ = $V_i$ - $V_{step}$
    **end if**
  **end for**
  Update the archive with respect to the obtained non-dominated solutions
**end for**

implementation of SetMOGWO with influence and maximin as its objectives is denoted by SetMOGWO_IM. Regardless of the number of objectives, we calculate all metrics. We model information propagation using the IC model and set the propagation probability to be $p = 0.1$ and seed set size to be $k = 25$ as in [7]. The population size is 500, and the number of iterations is 100. The results are averaged over 30 runs of the algorithms on all networks.

We examine 24 networks published by Wilder *et al.* [31] to model an obesity prevention intervention in the Antelope Valley region of California. Each node in a network is associated with demographic information: region, ethnicity, age, and gender. Each network has 500 nodes. Each demographic attribute partitions the network into two to 13 different groups. For each partitioning, all algorithms are compared. In [7] and also in our experiments, in each partitioning, each node belongs to a single group.

The characteristics of these networks, attribute values and group size for each value, are depicted in Table I. The values are averaged over all networks. For the region, as can be seen, there are 13 groups, 11 of which are smaller than 4% of the network. As expected, for most runs, values of all fairness objectives (speed, maximin, and rationality) for all algorithms on the region attribute became zero. Thus, we did not consider this attribute in our experiments.

TABLE I
DATASET CHARACTERISTICS

| Attributes | Attribute Values | Group Size (%) |
|---|---|---|
| Age | 18-24 | 14 |
| | 25-29 | 9.4 |
| | 30-39 | 22 |
| | 40-49 | 26 |
| | 50-59 | 12 |
| | 60-64 | 6.1 |
| | 65+ | 10 |
| Gender | Male | 48.7 |
| | Female | 51 |
| Ethnicity | Asian | 3.5 |
| | Black | 13 |
| | Latino | 29.7 |
| | Other | 4.16 |
| | White | 49.7 |
| Region | Acton | 2.2 |
| | Desert_view_highlands | 0.6 |
| | Lake_los_angeles | 3.6 |
| | Lancaster | 42.2 |
| | Leona_valley | 0.4 |
| | Littlerock | 0.4 |
| | Northeast_antelope_valley | 1.4 |
| | Northwest_antelope_valley | 1 |
| | Northwest_palmdale | 1 |
| | Palmdale | 39.6 |
| | Quartz_hill | 3 |
| | Southeast_antelope_valley | 1.6 |
| | Sun_village | 3 |

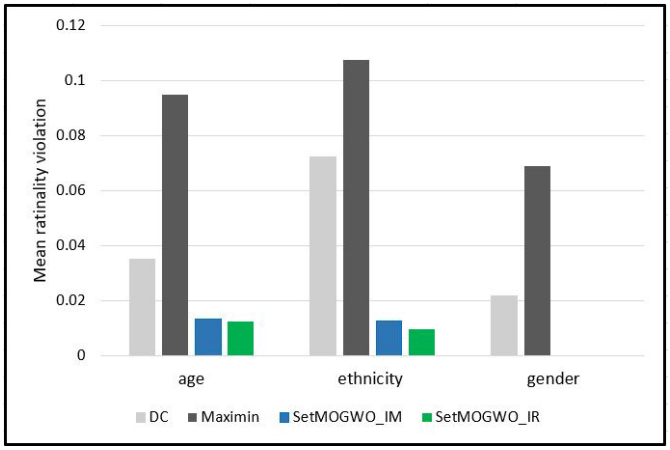
To evaluate the proposed algorithm, we compare our results with the two algorithms of the previous work (**Algorithm DC** to enforce rationality metric and **Algorithm Maximin** to enforce maximin metric) by Tsang *et al.* [7] and with the single-objective version of our proposed algorithm (called **SetSOGWO**). The source code of our algorithm is publicly available in [32].

SetMOGWO is an extension of our previous work (SetSOGWO [30]) for multiobjective problems. Experimental analysis showed that the size of the archive (the set of nondominated solutions) in all steps of the algorithm always remains small, resulting in constant time consumption for leader selection and archive management. Other parts of the two algorithms have the same time complexity. Therefore, the computational complexity of SetMOGWO is the same as SetSOGWO. In [30], experimental results on different real-world combinatorial optimization problems and datasets with different sizes show that SetSOGWO outperforms other existing methods (such as the genetic algorithm [35], PSO [36], artificial bee colony [37], firefly [38], and cuckoo search [39]) in terms of running time and scalability.

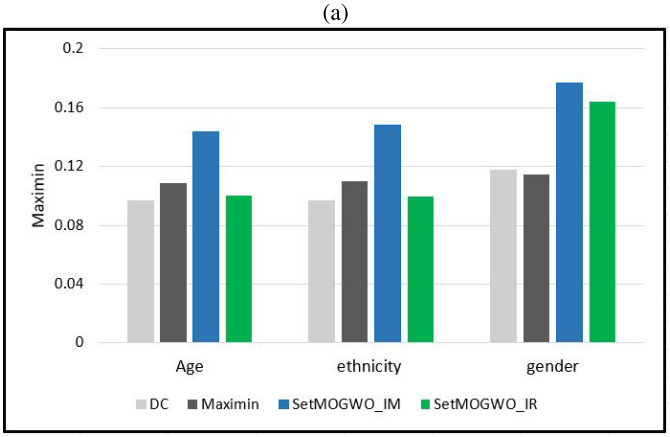
### A. Comparison With the Previous Work

We compare DC and Maximin with their corresponding SetMOGWO versions, that is, SetMOGWO_IR and SetMOGWO_IM. As in [7], for the group rationality metric, we evaluate the performance of algorithms by reporting the

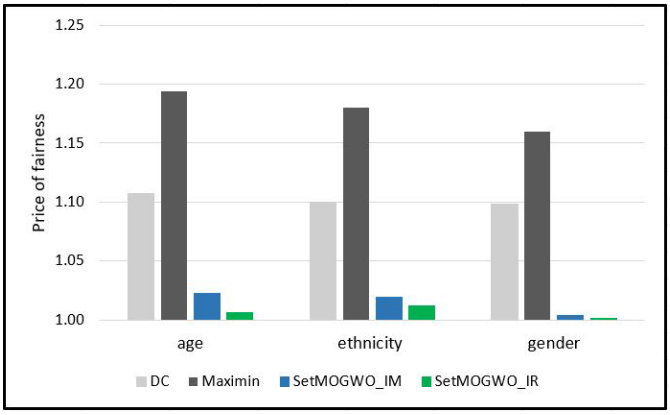
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(a)




(b)




(c)

Fig. 1. Comparison of SetMOGWO with the previous work [7] based on (a) rationality violation, (b) maximin, and (c) PoF.

mean percentage violation of diversity constraints over all groups. For other metrics, we directly report the mean value of the metric over all networks. Four mentioned algorithms are compared based on rationality violation, maximin, and PoF in Fig. 1.

As can be seen, SetMOGWO_IM outperforms other algorithms for the maximin metric, and SetMOGWO_IR outperforms other algorithms for the group rationality metric. Both SetMOGWO_IR and SetMOGWO_IM achieve zero rationality violation for the gender attribute because it partitions the network into two balanced groups. Also, both SetMOGWO_IM and SetMOGWO_IR achieve these performances by paying less price than DC and Maximin.

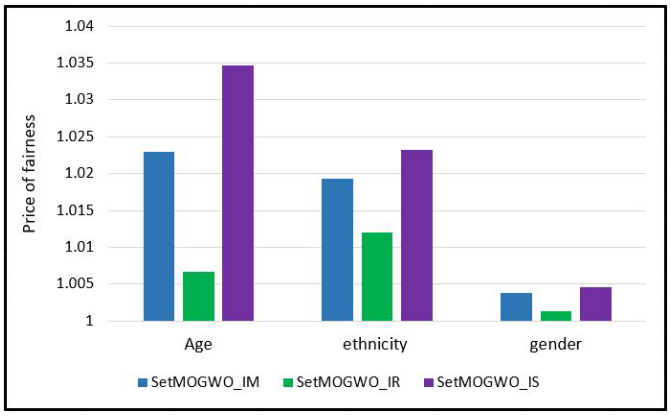



Fig. 2. Comparison of fairness metrics based on PoF.

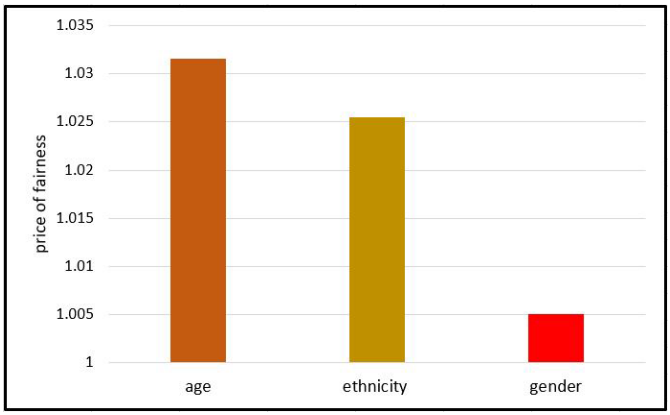



Fig. 3. Comparison of attributes based on PoF.

### *B. Which Group-Fairness Metric Has Less or More Price?*

A question that arises is that satisfying which fairness metric has less or more price than others? In Fig. 2, the prices of two-objective SetMOGWO versions are compared for different attributes. Expecting, the price of group rationality is lower than others because it does not try to maximize other fairness metrics. That is, a rational solution can have a low speed or maximin. On the other hand, the price of speed is higher than others because, for the speed metric to have a high value, maximin and rationality metrics must be satisfied.

### *C. Which Attribute Has Less or More Price?*

The results in Fig. 2 also show that the PoF for a specific metric is not the same for all attributes. To clarify more, in Fig. 3, the average prices of maximizing fairness metrics in two-objective SetMOGWO algorithms for different attributes are depicted.

As can be seen, the PoF for gender is less than others because it has only two balanced groups. As the number of groups and the imbalance among groups of an attribute increase, the PoF for that attribute also increases.

### *D. What Is the Effect of Selecting Different Attributes on Group-Fairness Metrics?*

We showed that, as the number of groups and the imbalance among them decrease, the PoF also decreases. Now, we study

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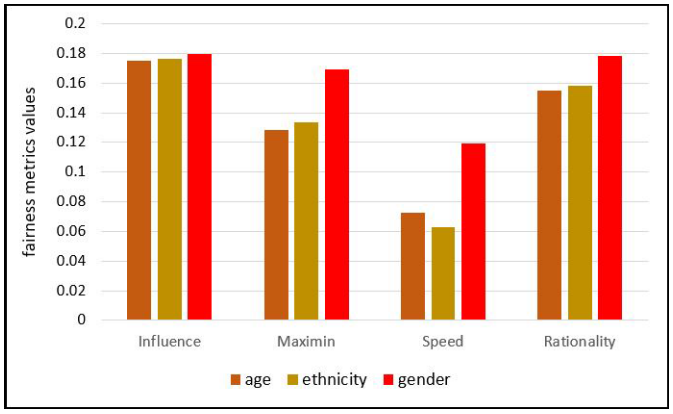



Fig. 4. Comparison of attributes based on the values of metrics.

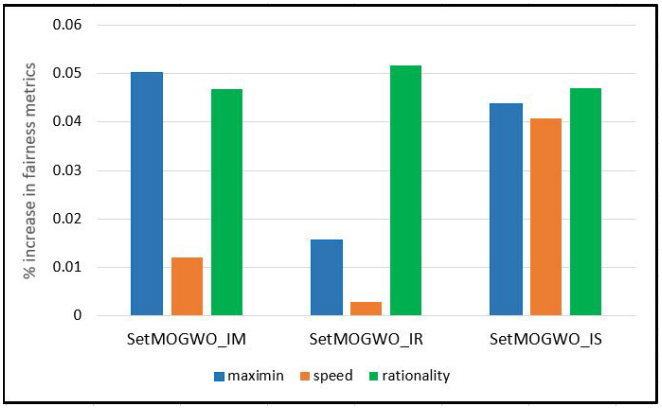



Fig. 5. Correlation between different fairness metrics.

the effect of selecting different attributes on fairness metrics. The average values of fairness metrics for all versions of SetMOGWO are depicted in Fig. 4. The results show that selecting different attributes can directly affect the metrics' values. As the number of groups and the imbalance among groups of an attribute decrease, the resulting values for all metrics increase. The amount of increase for influence is low, but, for the fairness metrics, it is considerable.

## VI. Correlation Between Fairness Metrics

By investigating the results, it can be found that there are meaningful relationships between some metrics. Fig. 5 depicts the extent of increase in fairness metrics in different two-objective versions of SetMOGOW. The results show that, as speed increases, rationality and maximin also increase. The reason is that, for the speed metric to be high, both rationality and maximin must be high; otherwise, the value of the speed metric cannot be increased. With the same argument, there is a meaningful relationship between maximin and rationality. By maximizing the minimum fraction influenced among all groups, we satisfy rationality but not vice versa.

### *A. How Does the Price of Fairness Change When the Number of Fairness Metrics Increases?*

We have considered two-objective IM problems, where the goal is to maximize the influence and one of the fairness metrics. But how does the PoF change when the goal

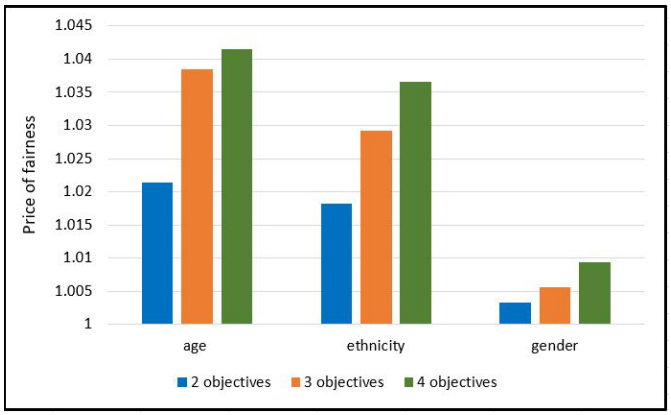



Fig. 6. Comparison of PoF based on the number of objectives.

is to maximize two or more fairness metrics? In Fig. 6, the average PoF for algorithms with different numbers of objectives is depicted. Expecting, as the number of fairness metrics increases, the price also increases. Interestingly, the PoF increases very slowly even for four-objective algorithms. For example, the PoF of the four-objective algorithm that tries to maximize influence and all three fairness metrics is at most 1.041. It indicates that satisfying fairness metrics is considerably less costly in real-world networks than the theoretical worst case would suggest.

### *B. Percentage Decrease in Influence Versus Increase in Fairness Metrics*

The PoF measures the price an algorithm pays for maximizing fairness metrics. However, it does not show the amount of increase in the fairness metrics. Fig. 7 depicts the percentage decrease in influence and increase in fairness objectives for all algorithms. The results show that a small decrease in influence can substantially improve other fairness metrics. For example, in SetMOGWO_IMRS, influence has decreased by 2.81%, but maximin, rationality, and speed have increased by 58.7%, 88.8%, and 49.5%, respectively.

### *C. Unfair Results by Ignoring Fairness Metrics*

Fig. 8 compares the single-objective algorithm (SetSOGWO) and two-objective algorithms for the age attribute based on the values of fairness metrics. The results show that SetSOGWO generates substantial unfair results. It generates the lowest satisfaction of group rationality and group activation speed, and has the smallest minimum fraction influenced. It seems that standard IM techniques can introduce substantial fairness violations. However, as stated before, reducing such violations is considerably less costly in real-world networks than the theoretical worst case would suggest. For example, in SetMOGWO_IS relative to SetSOGWO, influence has decreased by 3.3%, whereas maximin, rationality, and speed have increased by 60%, 78%, and 109%, respectively.

Fig. 9 has depicted the percentage increase/decrease in fairness metrics in SetMOGWO_IMRS compared to SetMOGWO_IMR. It shows that SetMOGWO_IMR ignores the order of nodes' activation, which can cause a disparity in resource allocation. On the other hand, using

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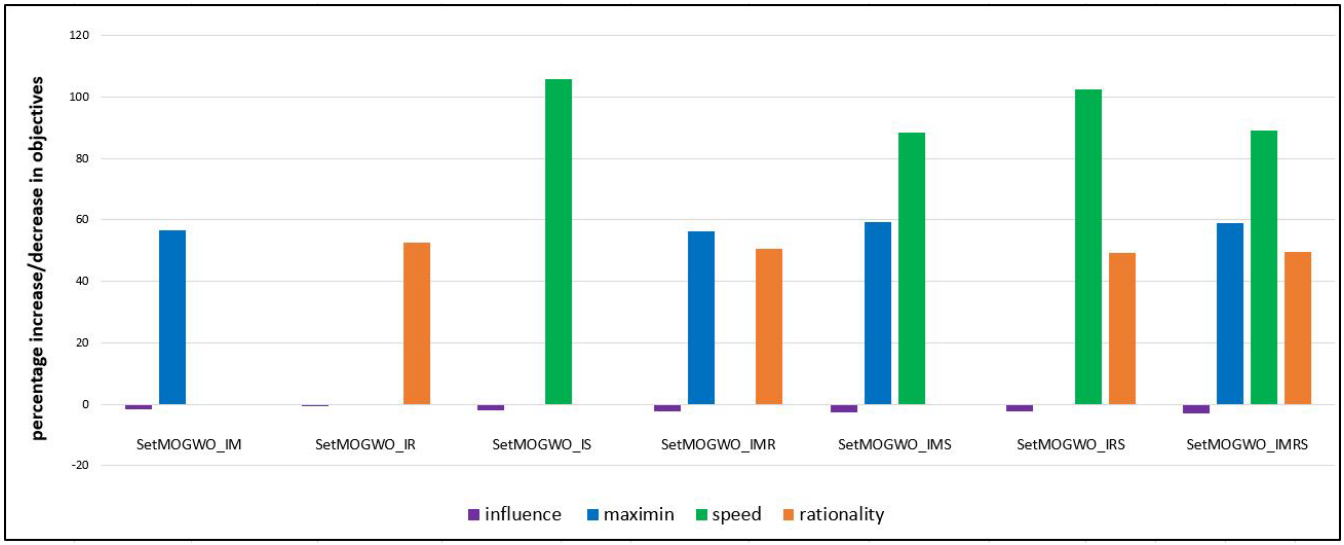



Fig. 7. Percentage decrease in influence versus increase in fairness objectives.

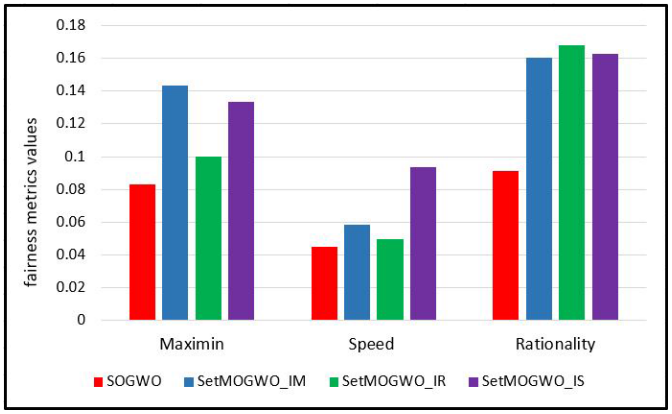



Fig. 8. Decrease in influence versus increase in fairness metrics for age attribute.

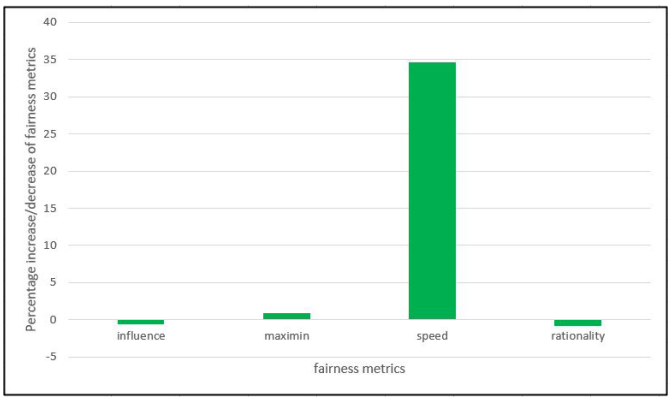



Fig. 9. Percentage increase/decrease in fairness metrics in SetMOGWO_IMRS compared to SetMOGWO_IMR.

SetMOGWO_IMRS can ensure that groups receive a fair portion of resources in a fairer order at relatively little cost.

## VII. Conclusion and Future Work

In this article, we studied group-fairness-aware IM, the problem of finding a seed set composed of $k$ nodes in a network to ensure the fair spread of information across all groups. It is a variation of the classic IM problem with additional fairness objectives.

We extended the notion of the group-fairness concept by considering the speed of information propagation in groups. Then, we proposed a discrete multiobjective optimizer (SetMOGWO), based on the MOGWO, to improve fairness metrics in group-fairness-aware IM. The results show that SetMOGWO outperforms the previous work. Also, we carried out a detailed experimental analysis of real-world networks and found interesting relationships between fairness concepts and network characteristics.

We showed that traditional IM techniques often neglect smaller groups. Using a multiobjective algorithm can ensure these groups receive a fair portion of resources in a fair order at relatively little cost. The results indicated that the speed metric has more price to satisfy. Also, we showed that the prices of fairness for attributes that partition the network into a few balanced groups are less than other attributes. Finally, although single objective algorithms generate substantial unfair results, the fairness metrics can be substantially improved with low costs using multiobjective algorithms, which will help ensure that small and marginalized groups are fairly treated.

The main advantage of the proposed algorithm over the previous work [7] is that SetMOGWO is a binary multiobjective algorithm, which is designed specifically for discrete optimization problems, such as IM. On the other hand, the previous work uses a continuous optimization subroutine, a Frank–Wolfe style algorithm [33], which is inherently designed for continuous optimization problems. Thus, their algorithm inevitably rounds the outputs to discrete sets as a mandatory step, which is time-consuming [34] and decreases the quality of solutions [30]. Also, the previous work is limited to multiobjective submodular optimization problems, whereas SetMOGWO is applicable to any discrete optimization problem.

There are several future directions for this research. First, we think that fairness in IM must be more studied because there is no general definition for fairness, and fairness metrics are application-dependent. Thus, different applications need

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different fairness metrics. Next, we plan to parallelize SetMOGWO to improve its scalability on larger datasets. Also, we can consider other multiobjective optimizers instead of SetMOGWO to maximize fairness metrics. Finally, we think that it is desirable to design a framework for group-fairness-aware IM, in which the tradeoff between different fairness metrics can be performed such that the framework can be customized for various applications.

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